

## Review article

# How to improve the attitude accuracy of the star sensor under dynamic conditions: A review

Liheng Ma<sup>a</sup>, Dongkai Dai<sup>b</sup>, Yuanman Ni<sup>c</sup>

<sup>a</sup> College of Weapons Engineering, Naval University of Engineering, No. 717, Jiefang Avenue, Qiaokou District, Wuhan, 430033, Hubei Province, China

<sup>b</sup> College of Advanced Interdisciplinary Studies, National University of Defense Technology, No. 109, Deya Road, Kaifu District, Changsha, 410073, Hunan Province, China

<sup>c</sup> Laboratory of Intelligent Control, Rocket Force University of Engineering, No. 2, Tongxin Road, Baqiao District, Xi'an, 710025, Shanxi Province, China

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## ABSTRACT

Star sensors experience significant degradation in attitude measurement accuracy under dynamic conditions, primarily caused by motion-induced star image blurring and diminished effective stars. This review analyzes the underlying mechanisms responsible for accuracy deterioration in dynamically perturbed star sensors. As the first comprehensive survey in this field, the paper proposes a novel taxonomy classifying existing mitigation strategies into two paradigm categories: active and passive deblurring techniques and motion blur compensation methods. We critically analyzed representative approaches within each category, including optical system optimization, multi-FOV configurations, servo stabilization platforms, and algorithm-driven solutions such as image restoration and attitude-correlated frame processing. A comparative evaluation highlights the advantages and limitations of these methods through quantitative performance metrics and implementation constraints. Furthermore, the study establishes selection criteria emphasizing that optimal strategy adoption must consider application-specific requirements. This work provides both a technical reference for star sensor designers and a framework guiding future research directions in high-dynamics attitude determination.

## 1. Introduction

The star sensor, as an absolute attitude measurement device, has gained extensive applications in spacecraft, satellites, ballistic missiles, and marine navigation systems due to its arc-second level precision [1–4]. Operating through stellar image acquisition via optical sensors (except for the Sun), it executes sequential procedures including star image preprocessing, star centroid, star identification, and attitude determination to establish spatial orientation from its body coordinate system with respect to the inertial celestial coordinate system.

The cross boresight (corresponding to the  $x$  and  $y$  axes) noise equivalent angle (NEA) error  $E_{ss}$  can be used to evaluate the attitude accuracy of the star sensor, and it is calculated as follows [5]:

$$E_{ss} = \frac{FOV_{1D} E_{centroid}}{n_{pixel} \sqrt{\bar{n}_{star}}} \quad (1)$$

where  $FOV_{1D}$  (degrees) denotes the field of view of the star sensor in one dimension,  $n_{pixel}$  (pixels) the sensor resolution,  $E_{centroid}$  (pixels) the average centroid accuracy, and  $\bar{n}_{star}$  the average detectable stars per frame.

For a certain star sensor,  $FOV_{1D}$  and  $n_{pixel}$  are constant; therefore, the attitude accuracy of the star sensor is mainly limited by the accuracy of the star centroid according to Eq. (1), and increasing the number of effective star images improves the attitude accuracy of the star sensor to a certain extent. However, under dynamic conditions, such as satellite orbital adjustments, spacecraft maneuvering, ship swinging, and other angular motion scenarios, there is angular motion around one or more axes of the star sensor, and star images become motion-blurred. The limited energy of the star image is dispersed into dozens or even tens of pixels. However, the noise in each pixel remains almost unchanged, leading to an abrupt decrease in the signal-to-noise ratio (SNR) of star image followed by star centroid accuracy degradation. On the other hand, some dim stars drown in noise, and thus the number of effective star images is reduced, as shown in Fig. 1 [6]. Fig. 1 compares star images under static (a) (star images are marked in red circles) and dynamic (b) (star images are marked in red rectangles and numbered) conditions, both of which were obtained under real night sky conditions at the same night and the exposure time interval between them was about two minutes, meaning that the number reduction of star image is aroused by the dynamic condition. The 2D energy distribution of a star

\* Corresponding author.

E-mail address: [jeremiahmax@163.com](mailto:jeremiahmax@163.com) (L. Ma).

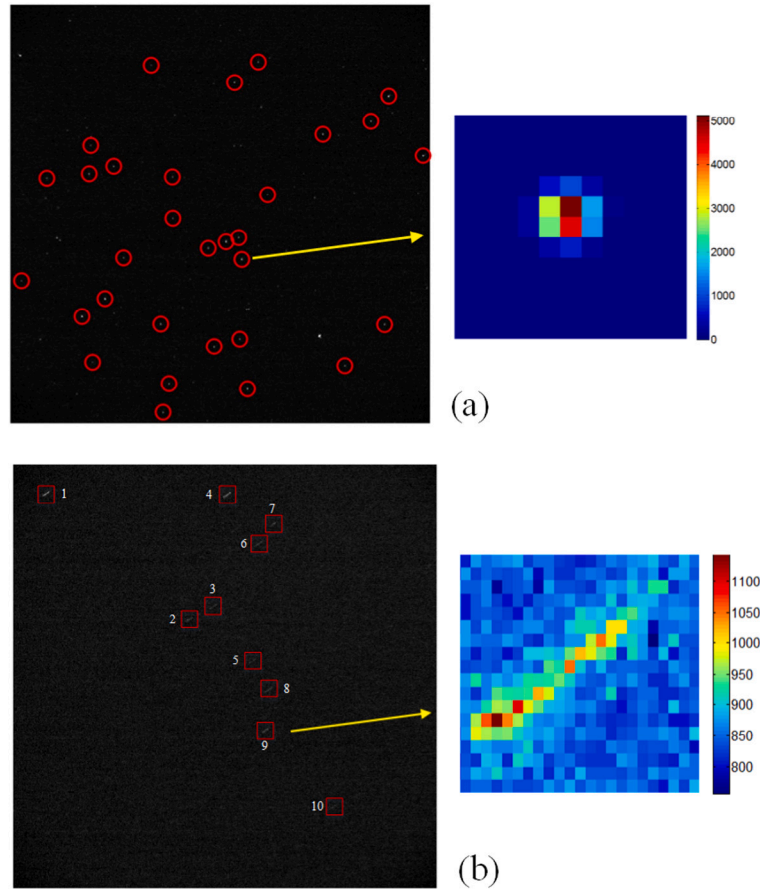


Fig. 1. Star images under static (a) and dynamic conditions (b). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

image in Fig. 1 is magnified and shown on the right side. A comparison of the two figures shows that the number of star images obtained under dynamic conditions is significantly reduced, the energy of the star images is dispersed into a dozen pixels, which significantly reduces the SNR. This dual degradation mechanism severely compromises both centroid precision and pattern recognition reliability, ultimately propagating attitude errors beyond mission tolerances.

Despite advances since pioneering star tracker designs in the 1990s [7], dynamic performance enhancement remains a persistent challenge. In this review, depending on whether star image motion blur is generated or not, existing methods are divided into two categories. Motion blur is caused by the relative angular motion between the star sensor and the star to be observed. Hence, to reduce the degree of motion blur of the star image or to suppress the formation of motion blur, the most direct method is to reduce the relative motion between them during the exposure time of the star sensor. A category of enhancement methods based on this principle is called active and passive deblurring methods (APDM) in this manuscript, which mainly includes improving the performance of the optical imaging system, multiple field-of-view (FOV) star sensor, servo-tracking platform techniques, dynamic binning algorithm (DBA), and time-delayed integration (TDI) method. Another category of methods is to compensate for the motion-blurred star image through the processing of software algorithm after the formation of motion blur, which is named the motion blur compensation methods (MBCM) in the review. MBCM includes motion-blurred star image restoration algorithms, star centroid with filtering method, and attitude-correlated frame (ACF) approach.

In this article, representative research on improving the attitude accuracy of the star sensor under dynamic conditions, including concept

ideas, principles, results, advantages, and disadvantages, is reviewed and summarized. The remainder of this paper is organized as follows. In Section 2, the APDM is reviewed and comparisons and summaries of different methods are presented. The MBCM is presented in Section 3. Finally, conclusions are drawn.

## 2. Active and passive deblurring methods

Active and passive deblurring methods (APDM) aims to minimize or suppress motion blur formation during the star sensor's imaging process. It is categorized as either active (requiring external actuation) or passive (inherent system optimization) based on operational principles in this review. Passive deblurring methods in this paper include techniques of improving the performance of optical imaging systems and multi-FOV star sensors, while active deblurring methods consist of techniques such as servo-tracking platform, DBA and TDI.

### 2.1. Improving the performance of optical imaging systems

Under dynamic conditions, the motion-blurred length of a star image can span tens of pixels, significantly exceeding the static point spread function (PSF). The blurring magnitude is directly proportional to the exposure time of the optical system of the star sensor, whereas the energy acquired per pixel depends on optical parameters as modeled by Eq. (2) [5]:

$$Photons \approx \frac{K \cdot \left( \int_{400}^{800} QE \cdot I(\lambda, T_k) d\lambda \right) \cdot S_{aperture} \cdot d}{f \cdot \omega \cdot 2.512 Mv} \quad (2)$$

where QE denotes wavelength-dependent quantum efficiency of the image sensor,  $I(\lambda, T)$  is the stellar energy spectral density distribution (400 nm to 800 nm in the visible band) which is related to the light wavelength  $\lambda$  in m and the stellar temperature  $T_k$  in Kelvin,  $K$  is the effective transmittance of the system (the atmospheric transmittance should be taken into account when on the ground),  $S_{aperture}$  is the aperture of the optical system,  $d$  is the pixel size of the image sensor,  $f$  is the focal length of the optical system,  $\omega$  is the equivalent angular velocity of the star sensor, and  $Mv$  is the star magnitude. Therefore, exposure time can be shortened by choosing an image sensor with a high QE, an optical lens with high transmittance, and an optical system with a large aperture. These are systematically and comprehensively summarized below:

#### A. Adaptive exposure time adjustment:

- **Ball HAST sensor** [7–9] employed large-aperture optics ( $f/1.4$ ) with back-illuminated CCDs (QE >80% @600 nm), achieving  $4^\circ/\text{s}$  operation by constraining blur length to <3 pixels through adaptive exposure time adjustment (1~10 ms). This introduces an inherent trade-off: the shorter exposure time reduces the number of detectable stars by 22% compared to the static modes [10].
- **Lockheed Martin's AST-301** [11] implemented similar variable exposure time control but prioritizes faint-star detection (limiting magnitude 6.5) through exposure time optimization algorithms.

#### B. High-sensitivity sensor implementation:

- **Electron multiplying charge coupled device (EMCCD):** Electron multiplication gain (>1000 $\times$ ) allows sub-ms exposures but induces multiplicative noise (Excess Noise Factor = 1.4~2.0). Experimental results [12] demonstrated 0.8 pixel centroid accuracy at  $8^\circ/\text{s}$ , albeit with 15% false star rates due to noise amplification.
- **Intensified charge coupled device (ICCD):** Microchannel plate intensifiers achieve 1 ms exposures ( $25^\circ/\text{s}$  tolerance), but suffer from limited dynamic range (54 dB vs. CCD's 72 dB) and radiation-induced gain decay (3% annual loss in LEO) [13–16].

**C. Optical parameter optimization:** Aperture scaling ( $D \propto \sqrt{SNR}$ ) improves sensitivity but exacerbates distortion ( $\epsilon_{dist} \propto D^3/f$ ). For instance, a 120 mm aperture induces 2.8 arcsec distortion versus 1.1 arcsec for 80 mm designs [8]. Field tests reveal optimal  $f/\#$  ranges (1.2~1.8) balancing irradiance and aberrations.

However, a comprehensive analysis and evaluation should be performed to select these optical parameters. For example, the star image distortion error increases simultaneously with the aperture. Meanwhile, despite theoretical advantages, implementation barriers persist:

#### A. Material constraints:

- Peak QE capped at 95% due to anti-reflection coating limitations.
- Fused silica substrates restrict  $D/f$  ratios to <0.25 for distortion control.

#### B. Cost considerations:

- EMCCD/ICCD systems incur several times cost premiums versus conventional CCDs.
- Large-aperture aspheric lenses increase manufacturing costs.

#### C. Performance plateaus:

- Optical enhancements alone yield <50% NEA improvement (HAST: 0.57 arcsecs  $\rightarrow$  0.42 arcsecs @ $3^\circ/\text{s}$ )
- Diminishing returns observed for QE >90% or  $f/\# < 1.0$

Case studies confirm that standalone optical optimization provides insufficient dynamic performance gains, necessitating hybrid approaches combining hardware upgrades with algorithmic compensation (see Section 3).

#### 2.2. Multi-FOV star sensors

The design of star sensors with multi-FOV aims to enhance attitude measurement reliability and dynamic performance through spatial redundancy and expanded sky coverage. A multi-FOV star sensor combining two or more FOVs into a single FOV offers several benefits. First, to improve the reliability of the attitude measurement of the star sensor, a redundant backup strategy is generally adopted; that is, one or two star sensors are added as backups [11,17], which is essential for applications of the star sensor in space. Second, multi-FOV are mounted at an angle to each other, and the measurement results from different FOVs are fused through accurate calibration among the FOVs to achieve high precision attitude information output, particularly for the attitude measurement accuracy of the boresight angle of the system, typically the angle around the z-axis [18]. Then, the orientation of the installation of the star sensors in the application needs to consider the influence of the sun, moon, and reflected light from the Earth [5]. To overcome this influencing factor, a scheme can be used that combines multi-FOV facing different sky zones. Finally, the number of effective star images increases with the number of FOV. As the number of star images increases, the exposure time can be appropriately reduced, and the effect of the reduction in the number of star images in a single FOV due to the motion of the carrier can be partially eliminated, which also improves the attitude accuracy of the star sensor under dynamic conditions.

This section analyzes four representative implementations of multi-FOV star sensors, detailing their operational principles, advantages, and limitations.

##### A. Tri-optical-head configuration: HYDRA System

Developed by SODERN (France), the HYDRA star sensor [11] integrated three independent optical modules arranged at  $120^\circ$  intervals (Fig. 2). Each module employs an active pixel sensor (APS) with a  $22^\circ \times 22^\circ$  FOV, achieving a multi-FOV of  $68^\circ \times 22^\circ$ . Key performance characteristics include:

- **Attitude accuracy improvement:** Cross-boresight NEA reduced from 17.4 arcsecs (x/y axes) and 145 arcsecs (z-axis) in single-FOV mode to 6.7 arcsecs (all axes) under  $5^\circ/\text{s}$  dynamic condition.
- **Dynamic tolerance:** Recognizable star images maintained at angular velocities up to  $10^\circ/\text{s}$  through real-time data fusion.
- **Hardware trade-offs:**
  - Mass increased by 210% 4.2 kg vs. 2.0 kg for single-FOV systems
  - Power consumption escalated to 28 W (vs. 10 W baseline)
  - Calibration complexity grew quadratically, requiring 12 degrees of freedom (DOF) alignment versus 3 DOF for single-FOV configurations [18].

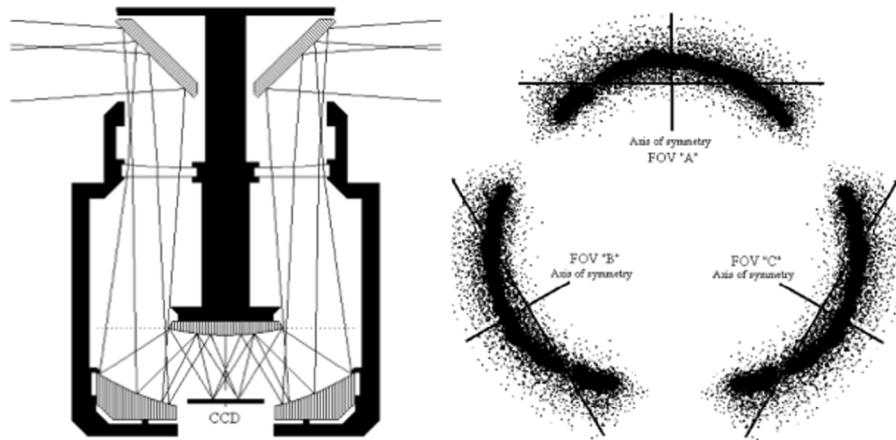


Fig. 2. Three FOVs star tracker — StarNav III [11].

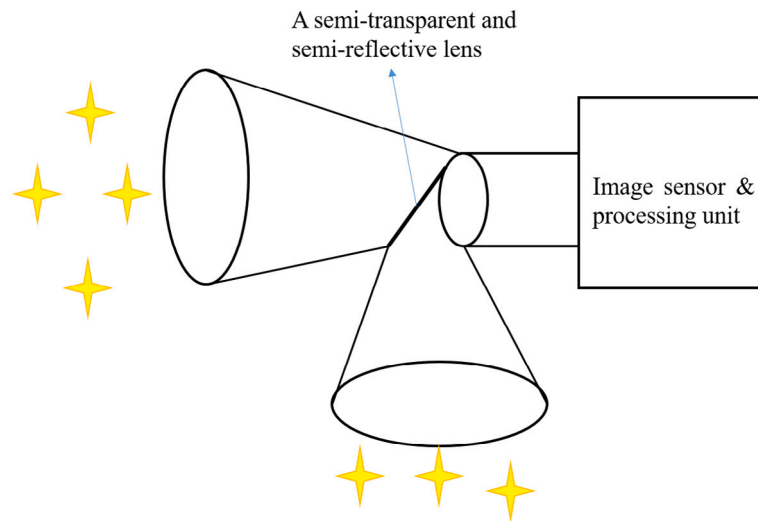


Fig. 3. Two FOVs star sensor.

#### B. Reflective multi-FOV architecture: ASI/Galileo spa collaboration

Funded by the Italian Space Agency (ASI), this design [17] uses dichroic mirrors to project three FOVs onto a single CCD through a folded optical path:

- **Advantages:**
  - 40% cost reduction compared to multi-head systems
  - 18% improvement in boresight (z-axis) measurement accuracy through redundant observations
- **Challenges:**
  - 35% irradiance loss per reflection path due to mirror coatings
  - Stray light contamination between FOVs caused 12% SNR degradation in lab tests
  - Dynamic performance remains unquantified due to limited prototype testing

#### C. Semi-transmissive dual-FOV design: Tsinghua University prototype

This configuration (Fig. 3) employed beam splitters to simultaneously image two  $14^\circ \times 14^\circ$  FOVs on a single detector:

- **Performance benefits:**
  - 55% FOV overlap enabled cross-validation of star patterns, improving identification success rate by 22%
  - Static accuracy reached 3.1 arcsecs under laboratory conditions
- **Operational limitations:**
  - 50% light attenuation in each optical path degraded SNR by 6 dB
  - Earth albedo interference in one FOV contaminated the other FOV, inducing >15% false star detections
  - Dynamic accuracy deteriorated to 8.9 arcsecs under  $3^\circ/s$  motion [19]

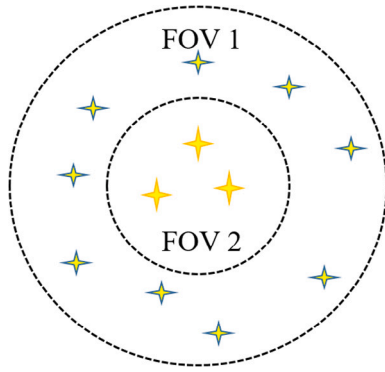


Fig. 4. A finer small FOV within the center of a large FOV.

#### D. Nested FOV architecture: Huazhong University approach

Developed by Huazhong University of Science and Technology [20], this hierarchical design (Fig. 4) combines:

- **Wide-FOV subsystem:**  $8^\circ \times 8^\circ$  coverage with  $512 \times 512$  pixels (3.5 arcsecs per pixel resolution) for rapid initial star identification (<200 ms)
- **Narrow-FOV subsystem:**  $0.5^\circ \times 0.5^\circ$  coverage with  $1024 \times 1024$  pixels (0.17 arcsecs per pixel resolution) for high-precision centroiding (0.02 pixel accuracy)
- **System constraints:**
  - Limited to low-dynamic conditions ( $<0.5^\circ/\text{s}$ ) due to PSF mismatch between FOVs
  - Data fusion latency increased by 40% compared to single-FOV operation
  - No dynamic performance validation reported in accessible literature

#### E. Comparative analysis and implementation challenges

While multi-FOV designs theoretically enhance reliability, practical deployment faces critical hurdles:

- **Calibration complexity:** Tri-head systems require 12-parameter alignment matrices versus 5 parameters for dual-FOV configurations [18].
- **Optical performance degradation:**
  - 1 dB SNR loss per additional optical path component (lenses/mirrors)
  - 15%–25% modulation transfer function MTF reduction in wide-FOV optics
- **Dynamic limitations:**
  - Maximum validated angular rate remains  $10^\circ/\text{s}$  (HYDRA) versus  $3^\circ/\text{s}$  for semi-transmissive designs
  - Latency penalties of 33 ms per additional FOV due to parallel processing

Although spatial redundancy enhances fault tolerance in these implementations, the most efficient is the HYDRA star sensor [11] with three optical heads. The improvement in attitude accuracy of other multi-FOV star sensors [17,19,20] is neither impressive nor reported further. Meanwhile, the associated trade-offs — increased mass, elevated power demands, and complex calibration requirements — mandate rigorous evaluation tailored to specific operational contexts.

#### 2.3. Servo tracking platform techniques

To mitigate relative motion between the star sensor and its carrier, some star sensor products have adopted a servo tracking platform positioned between them [18]. This servo mechanism dynamically tracks the carrier's motion in real time, maintaining a fixed pointing direction for the star sensor. Consequently, the relative motion between the star sensor and observed stars is effectively eliminated, thereby preventing motion-blurred star image formation. Such servo tracking platforms are predominantly integrated into navigation systems. For instance, Northrop Corp. has implemented this technology in its NAS-27 and LN-120G INS/CNS navigation systems [21].

Beyond Northrop Corp., the China Huazhong Institute of Optoelectronic Technology has also explored servo tracking platforms [22], however, the available information is extremely scarce. In [23], the servo mechanism has been used to drive the precise azimuth axis pointing the optical axis of the star sensor to the star. A fine star image centroid extraction performance with an extraction error of less than 0.2 pixels or 2 arc-second under four different types of one-dimensional angular motion after compensation has been demonstrated. Nevertheless, published literature on these efforts remains scarce, limiting detailed technical evaluations of their implementations.

Under dynamic conditions, the servo mechanism ensures smooth operation of the star sensor. However, inherent error factors — such as axis system misalignments and encoder inaccuracies — degrade both attitude measurement precision and system reliability. Furthermore, the servo platform's large size, structural complexity, and high cost render it unsuitable for mass- and size-constrained applications like satellites. Instead, it finds greater suitability for ship-based platforms where these constraints are less stringent.

#### 2.4. DBA and TDI method

Researchers at the European Space Agency proposed a conceptual framework for the dynamic binning algorithm (DBA) tailored to APS star sensors [24]. The DBA compensates for motion-induced distortions by combining pixel readouts from consecutively sub-sampled images via an oversampling technique. This process relies on three-axis motion parameters obtained from a low-precision gyroscopic unit. However, the DBA method is limited to compensating motion along horizontal, vertical, and pixel diagonal directions. For other diagonal trajectories, motion must be decomposed into combinations of these predefined directions, as illustrated in Fig. 5. This decomposition approach renders the method ineffective for addressing complex multi-directional motion patterns.

Notably, the DBA requires specialized logic circuits for APS sensors to achieve virtual charge accumulation at the pixel level. While theoretically promising, its practical implementation remains unverified, and no subsequent reports or validated applications have emerged in accessible literature.

Similar to the DBA, the time-delayed integration (TDI) method [25] synchronizes photogenerated charge movement with star image motion through customized drive timing circuits. In CCD imaging arrays, charges generated by starlight follow the carrier's motion along the  $y$ -axis, ensuring overlap in the  $y$ -direction. This synchronization enhances the signal-to-noise ratio (SNR) when paired with specialized image processing algorithms.



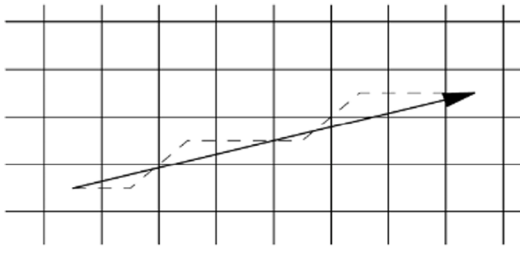


Fig. 5. Motion decomposition along diagonal direction with the DBA method.

The TDI technique has been implemented in Lockheed Martin Corp.'s AST-201 [25] and AST-301 [7] star sensors. For example, the AST-301 sensor, deployed on the Space Infrared Telescope (SIRTF) at the Jet Propulsion Laboratory (Fig. 6), eliminates  $y$ -axis motion blur while retaining  $x$ -axis blur. In static conditions, this sensor achieves attitude accuracies ( $1\sigma$ ) of 0.18, 0.18, and 5.1 arcsecs for the  $x$ -,  $y$ -, and  $z$ -axes, respectively. Under dynamic scenarios, it maintains a hold accuracy of 1 arcsec/s and a continuous tracking accuracy of 2.1 arcsec/s. Studies at the Technical University of Denmark further validated the TDI method's efficacy in spacecraft rotational motion tracking [8].

Despite these advantages, the TDI method increases the complexity of hardware timing circuits. Furthermore, the charge-transfer region formed during operation reduces the effective imaging area of the CCD. These limitations restrict its applicability to large-format frame-shift CCD arrays and render it unsuitable for handling intricate multi-axis motion scenarios.

### 3. Motion blur compensation methods

Active and passive de-blurring methods tend to add extra hardware costs and increase system complexity. In recent years, there have been more studies on motion blur compensation algorithms for star images. The motion-blurred star image compensation method compensates for the motion-blurred star image by designing relevant algorithms after the formation of star image blurring to improve the precision of star centroid. Motion blur compensation methods include motion-blurred star image restoration algorithms, star centroid with filtering algorithms, the attitude-correlated frame approach, and the attitude-correlated frame adding method. Before specifically introducing the related compensation algorithms, it is necessary to review the related research work on the motion-blurred model of star images and summarize the related methods of motion-blurred star image preprocessing.

#### 3.1. Modeling of star image motion blur

To effectively mitigate motion blur through algorithmic compensation, establishing an accurate mathematical model of star image motion blur under dynamic conditions is essential. Assuming the relative motion between the star sensor and the observed star is  $(x_0(t), y_0(t))$ , the motion-blurred image  $g(x, y)$  can be expressed as:

$$g(x, y) = \int_0^T f(x - x_0(t), y - y_0(t)) dt + n(x, y), \quad (3)$$

where  $T$  denotes the exposure time of the star sensor,  $f(x, y)$  represents the original unblurred star image, and  $n(x, y)$  accounts for additive noise. Applying the Fourier transform to Eq. (3), the relationship in the frequency domain becomes:

$$G(u, v) = F(u, v) \int_0^T e^{j2\pi(ux_0(t)+vy_0(t))} dt + N(u, v), \quad (4)$$

where  $G(u, v)$  and  $F(u, v)$  are the Fourier transform of  $g(x, y)$  and  $f(x, y)$ , respectively. The motion-blurred degradation function in the frequency domain is thus derived as:

$$H(u, v) = \int_0^T e^{j2\pi(ux_0(t)+vy_0(t))} dt. \quad (5)$$

This model clarifies that, assuming precise noise removal and accurate estimation of motion parameters, the original unblurred star image can theoretically be reconstructed from the degraded observations. The degradation and restoration process is conceptually illustrated in Fig. 7.

Research on motion-blurred star image modeling has been extensively conducted by Beihang University. In [26], the dynamic performance of a star sensor was simulated by incorporating motion functions into static imaging models. This approach enabled the establishment of a dynamic star image model and subsequent validation of restoration algorithms through simulations. Further analysis of error sources and system parameter optimization was also performed. The work in [27] systematically investigated the mechanisms of star image blurring under various carrier motion types (e.g., linear, rotational) and developed corresponding blurring models. These models were validated through simulations, demonstrating their effectiveness in guiding restoration algorithms.

A comprehensive study in [28] integrated rotational motion around three axes to derive equations describing star image trajectories. Simulations confirmed the accuracy of these models under complex rotational dynamics. In [29], the energy distribution of motion-blurred star images was modeled by linearly segmenting the motion-blurred degradation function. This model incorporated critical imaging parameters such as incident light intensity, exposure time, star image velocity, and Gaussian radius. An analytical expression for centroid error was derived, enabling precise error evaluation and parameter optimization. Both simulation and experimental data validated the model's accuracy.

For missile-borne star sensors, [30] established an energy distribution function for uniformly moving star images based on kinematic equations. Two solutions were proposed to estimate star image velocity, and the spatial variation of motion functions across the image plane was addressed by including nonlinear terms. This refined model provided a foundation for evaluating the precision of compensation algorithms [31]. Complex scenarios involving variable acceleration were further explored in [32], with similar studies reported in [33–35]. A simulated motion-blurred star image generated using these models is shown in Fig. 8(a), while Fig. 8(b) demonstrates strong correlation between simulated and real motion-blurred star images [6].

Collectively, these studies have advanced the understanding of motion-blurred star image formation mechanisms. Models accounting for uniform, accelerated, and variable-velocity motions provide critical insights for subsequent processing algorithms, such as star image restoration and centroid accuracy enhancement.

#### 3.2. Preprocessing of motion-blurred star image

To improve the SNR and centroid accuracy of motion-blurred star images, preprocessing plays a critical role after the formation of motion blur. As indicated by Eqs. (1) and (4), the primary objective of preprocessing is to eliminate noise components from the degraded star images while preserving essential features for subsequent processing steps.

In [36], a highly dynamic star image extraction method was proposed, where an adaptive processing window was dynamically adjusted based on the motion characteristics of the star sensor. This adaptive window accommodated diverse motion patterns, enabling robust star image extraction within localized regions. For fragmented star images caused by rapid motion, mathematical morphology techniques were employed to detect breakpoints and actively "grow" the broken segments, ensuring continuity. Additionally, wavelet transforms were utilized to denoise motion-blurred star images [37], effectively separating noise from structural features in the frequency domain.

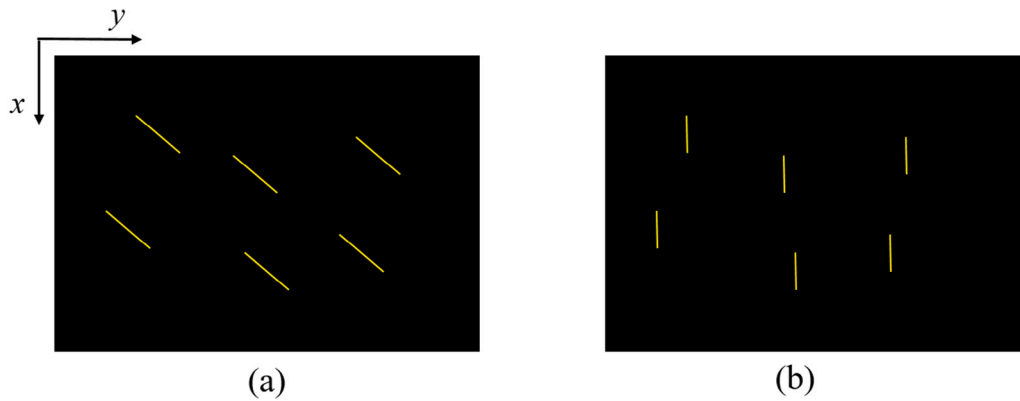


Fig. 6. TDI for SIRTf star tracker. The motion in the  $y$  direction (a) is eliminated with the TDI in (b).

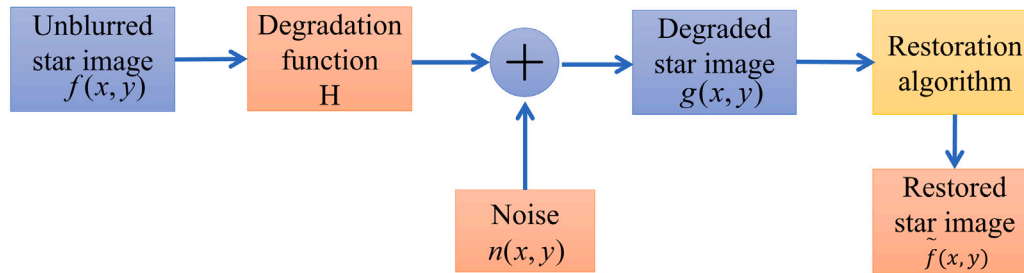


Fig. 7. Diagram of star image degradation and restoration.

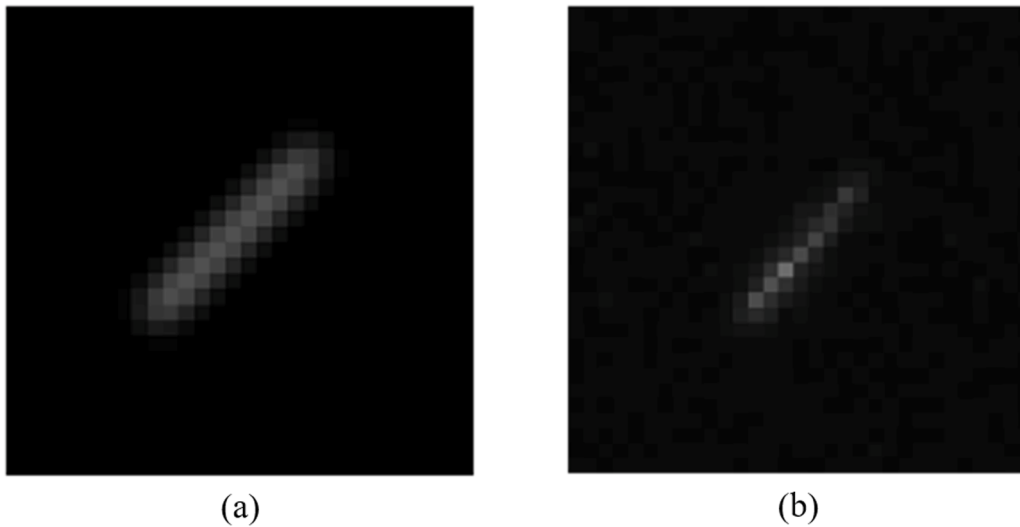


Fig. 8. Motion-blurred star image simulation (a) and a real motion-blurred star image (b).

Sun et al. [38] introduced a multi-stage preprocessing framework. First, correlation filtering was applied to suppress noise and enhance the SNR of motion-blurred star images. Second, background noise was extracted and subtracted using morphological opening operations, facilitating accurate segmentation of star image regions. The Sobel operator was then employed to compute gradients, and autocorrelation analysis of the differential images yielded estimates of motion parameters (e.g., direction and blur length). This approach enabled precise localization of motion-blurred star regions for subsequent extraction.

Wan et al. [39] proposed a method named optimal directional connected component (ODCC). According to the dynamic characteristics of star sensors, the ODCC enhancement method can adaptively estimate the directions of star spots and integrate the star image so that the SNR of the star spots is increased. It was used in its multi-FOV star sensor

to improve the faint star spot extraction ability under extremely high dynamic conditions [40].

Our research group has investigated various preprocessing techniques for motion-blurred star images. In [41], correlation matching was adopted to align and enhance star patterns under dynamic conditions. Mathematical morphology operations, as detailed in [42], were applied to repair fragmented star images (see Fig. 9(a)–(b)), where broken regions (marked by dashed red ellipses) were reconstructed through dilation and erosion processes. Furthermore, a star-seed pixel growth algorithm combined with region restriction [43] was developed to isolate valid star pixels from noisy backgrounds, significantly improving SNR. With these methods, the motion-blurred star images were well preprocessed prior restoration further in [4,43,44].

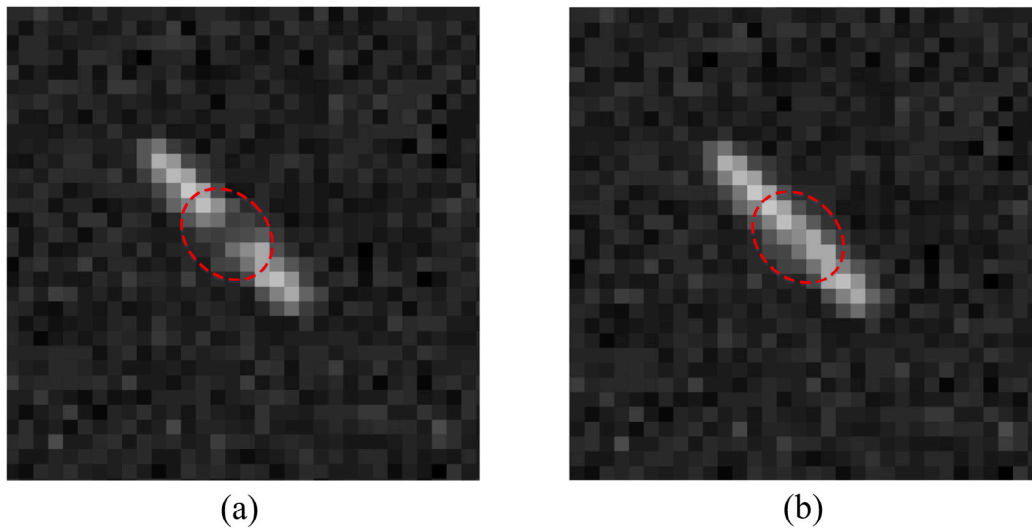


Fig. 9. Broken motion-blurred star image (a) after repairing (b).

Preprocessing of motion-blurred star images is indispensable prior to star centroid. Given the inherently low SNR under dynamic conditions, these steps are crucial for preparing high-quality inputs for downstream tasks such as image restoration, centroid estimation, and attitude determination.

### 3.3. Restoration of motion-blurred star image

The drastic reduction in SNR of motion-blurred star images, caused by energy dispersion across multiple pixels, necessitates advanced restoration techniques. The image restoration method attempts to regroup the dispersed energy of the star images into localized regions (e.g.,  $3 \times 3$  or  $5 \times 5$  pixel windows) to improve the SNR of the star images.

Jin et al. [36] employed Wiener filtering for motion-blurred star image restoration. As a linear minimum mean square error estimator, Wiener filtering effectively suppresses noise but introduces artifacts such as the ringing effect near sharp edges. While this method improves SNR to some extent, its limited noise robustness and residual edge errors constrain its practical utility for high-precision star centroid.

In contrast, Sun et al. [34] leveraged MEMS gyroscope-derived motion parameters to guide the Richardson–Lucy algorithm (RLA), a nonlinear iterative restoration method. RLA's inherent noise suppression capability and adaptability to PSF variations yielded significant improvements in centroid accuracy. For instance, under angular velocities of  $8^\circ/\text{s}$ , RLA reduced centroid errors from 1.8 pixels to sub-pixel levels. To further enhance performance, our group proposed a region-restricted RLA [43], which confines the restoration process to predefined regions of interest containing valid star signatures. As illustrated in Fig. 10(a) and (b), this approach achieves concentrated energy redistribution and markedly improved SNR. Fig. 10(a) shows raw motion-blurred star images, while Fig. 10(b) demonstrates restored images with concentrated features and improved SNR.

A critical requirement for most of the restoration algorithms is the accurate estimation of the motion-blurred degradation function  $H(u,v)$ , which depends on precise motion parameter acquisition. These parameters can be obtained via two approaches:

- **Image processing algorithms:** Techniques such as autocorrelation analysis [38] or gradient-based motion estimation [34] derive motion parameters directly from blurred star images.
- **Strapdown gyroscope integration:** High-precision gyroscopes [24,34,43] provide real-time angular velocity measurements to compute motion trajectories.

The motion-blurred star image restoration method can eliminate the influence of motion blur to a certain extent and improve the accuracy of the star centroids for star sensors. Despite their effectiveness, motion-blurred star image restoration methods face two key limitations:

- **Parameter sensitivity:** Inaccurate motion parameter estimation (e.g., due to gyroscope drift or algorithmic errors) degrades restoration quality.
- **Computational complexity:** Iterative algorithms like RLA increase processing latency, potentially affecting the update rate of attitude determination for the star sensor in real-time applications.

### 3.4. Star centroid with filtering algorithm

Motion-blurred star image restoration aims to improve the SNR of degraded star images, thereby enhancing centroid accuracy and ultimately boosting the attitude determination precision of star sensors under dynamic conditions. A promising strategy involves integrating filtering algorithms with predicted star positions derived from theoretical models, enabling refined centroid estimation even in challenging motion scenarios. Fig. 11 outlines the workflow of star centroid with Kalman filtering.

In [45], the extended Kalman filter (EKF) was employed to fuse angular velocity measurements from a strapdown gyroscope with star centroids obtained via the center-of-mass method. This fusion framework generated optimal estimates of star centroids under dynamic conditions, achieving an accuracy better than 1 pixel for stars of magnitude 5 under angular velocities of  $8^\circ/\text{s}$ . Liu et al. [46] utilized a Kalman filter to predict star positions; however, their work focused on static or mildly dynamic scenarios without addressing motion-blurred star images. Another study [47] proposed a directional integration method to enhance SNR by accumulating star energy along motion trajectories, where motion parameters were extracted directly from image processing algorithms. In [48], a generalized maneuvering attitude error-state Kalman filter (GMAESKF) was proposed to directly mitigate the attitude measurement noise and fulfill the requirement of angular velocity estimation for terrestrial multi-FOV star sensors when they are applied to agile maneuvering carriers. In the numerical simulation, the proposed method reduced the attitude measurement error by 42.3%, with an rms error of  $0.0171^\circ/\text{s}$  for angular velocity estimation. Flight experiment was carried out and verified the performance of the GMAESKF.



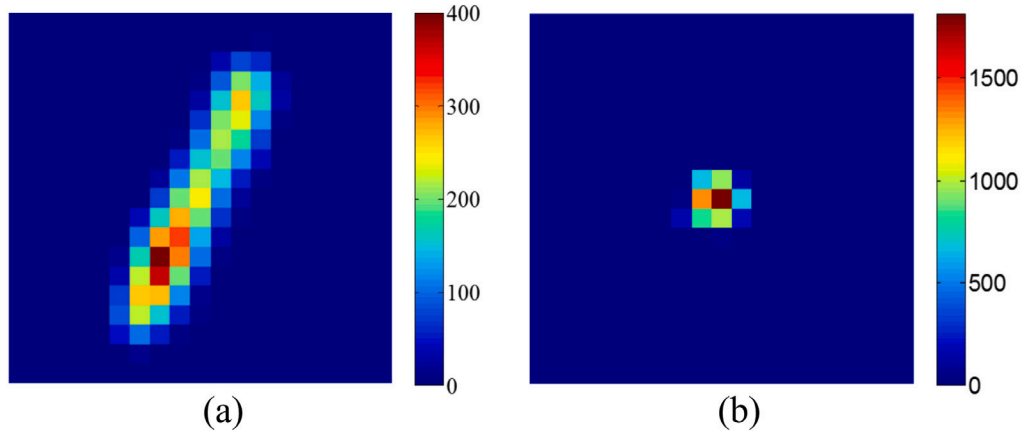


Fig. 10. Motion-blurred star image before (a) and after (b) employing the RLA restoration algorithm.

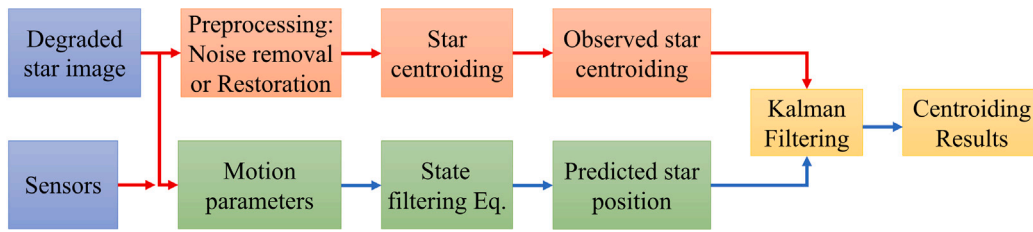


Fig. 11. Diagram of star centroid with Kalman filtering.

The Boeing Company further advanced this approach in a patented method [49]. A Kalman filter was designed to bridge the measured centroids (from the center-of-mass method) and predicted star positions (projected from the star catalog). The gyroscope unit provided two critical inputs:

- **Window localization:** Delineating the region of interest (ROI) for centroid calculation.
- **Initial attitude estimation:** Enabling star catalog projections for positional predictions. Simulation results demonstrated that this method reduced centroid errors from 1.8 pixels to 1/1000 pixels under non-uniform motion conditions. However, its performance hinges on two stringent requirements:
  - High-precision calibration of the mounting misalignment between the star sensor and gyroscope.
  - Ultra-accurate angular velocity measurements from the gyroscope unit.

Similar to star image restoration algorithms, the success of star centroid filtering methods relies heavily on precise motion parameter acquisition. These parameters can be sourced either from external sensors (e.g., gyroscopes) or derived algorithmically from image data, as illustrated in Fig. 11. While these methods significantly improve centroid accuracy, their practical implementation demands careful consideration of sensor calibration, computational efficiency, and real-time performance.

### 3.5. Attitude-correlated frame approaches

While the aforementioned motion blur compensation algorithms improve attitude accuracy under dynamic conditions to some extent, they remain constrained by the limitations of single-frame star image

processing. To overcome this, our research group proposed the attitude-correlated frame (ACF) method [35,44], which breaks through the limitation of the traditional method of single-frame star images, extends the thinking to the sequence of star image frames. By leveraging precise attitude transformations measured by a strapdown gyroscope unit, the ACF method correlates independent star image frames to virtually broaden the FOV and suppress random noise, thereby enhancing dynamic performance.

Fig. 12 illustrates the two-frame correlation process in the ACF approach. Star image frames captured at different times (e.g., Frame 1 at Time 1 and Frame 2 at Time 2) are correlated using the attitude transformation matrix provided by the gyroscope. This correlation effectively synthesizes a “virtual FOV” by combining star observations across multiple frames, analogous to the multi-FOV star sensor design described in Section 2.2. The improvement in attitude accuracy can be approximately quantified using the modified NEA formula in Eq. (6):

$$E_{ss} = \frac{FOV_{1D}}{n_{pixel}} \frac{E_{centroid}}{\sqrt{N \cdot \bar{n}_{star}}} \quad (6)$$

where  $N$  denotes the number of correlated frames. Both simulations and experiments confirm that the ACF method significantly enhances attitude accuracy of the star sensor under high dynamic conditions, with improvements approximately proportional to  $\sqrt{N}$ . For instance, as shown in Fig. 13(a), the attitude error decreases monotonically as  $N$  increases. Experimental results in Fig. 13(b) further validate this trend under angular velocities of 3.3°/s (z-axis), 1.8°/s (y-axis), and 0.17°/s (x-axis).

The ACF approach integrates multiple advanced techniques:

- **Single-frame restoration:** Motion-blurred star images in individual frames are preprocessed using the Richardson–Lucy algorithm (RLA) to improve SNR.
- **Centroid error compensation:** Systematic errors in centroid estimation are mitigated through calibration methods [50,51].

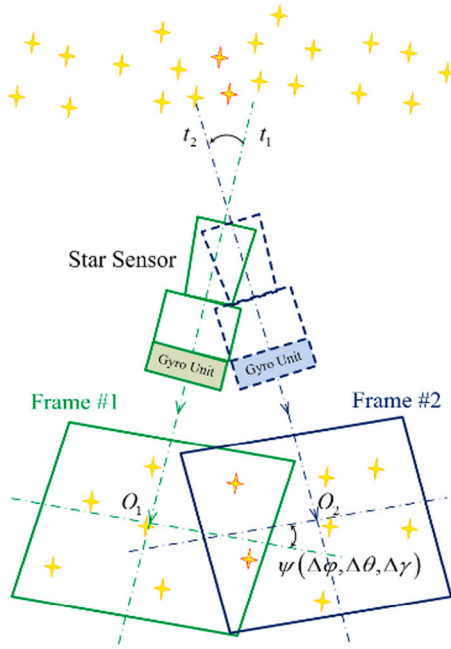


Fig. 12. Schematic diagram of two-frame correlation.

- **Sensor calibration:** Precise calibration of the installation misalignment between the star sensor and gyroscope [52,53] ensures accurate attitude transformation.
- **Internal parameter calibration:** Star sensor parameters (e.g., focal length, distortion coefficients) are optimized to minimize model mismatches [54,55].

A derivative approach, the attitude-correlated frame adding (ACFA) method [56], further enhances SNR by directly co-adding star images from correlated frames. Unlike ACF, which correlates star observation vectors, ACFA aligns and sums pixel-level data from multiple frames using gyroscope-measured attitude transformations. The accuracy improvement similarly follows the  $\sqrt{N}$  proportionality, demonstrating robustness under extreme dynamics.

Compared to hardware-based multi-FOV star sensors (Section 2.2), the ACF method achieves similar performance through time-division multiplexing rather than spatial FOV expansion. Key advantages include:

- **Noise suppression:** Random noise is averaged out across correlated frames.
- **Hardware simplicity:** Eliminates the need for complex multi-FOV optical systems.
- **Adaptability:** Compatible with existing strapdown gyroscope-star sensor architectures.

The efficacy of the ACF method has been extensively validated through simulations and field experiments. For example, in [52,53], the method demonstrated sub-arcsecond attitude accuracy under angular rates exceeding  $5^\circ/\text{s}$ . Applications span spacecraft, ballistic missiles, and shipborne platforms, where dynamic conditions are prevalent. However, the ACF method requires high-precision gyroscopes and rigorous calibration to ensure attitude transformation accuracy. Performance degradation may occur under rapid angular acceleration or gyroscope drift.

#### 4. Conclusion

The declining attitude accuracy of star sensors under dynamic conditions, primarily caused by motion-induced star image blurring

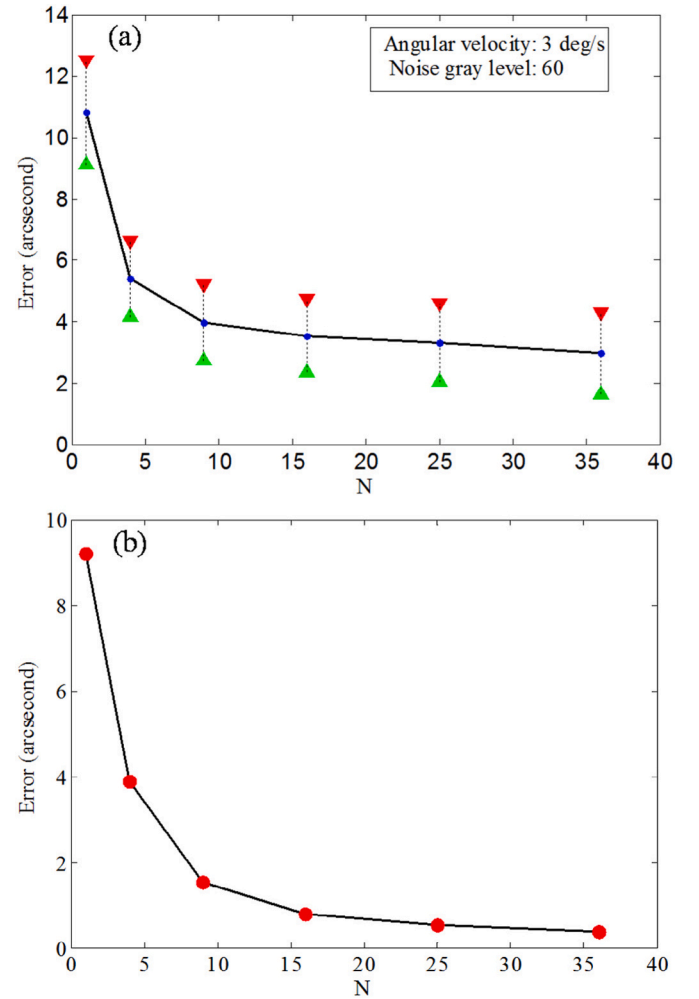


Fig. 13. Simulation (a) and experiment (b) results with ACF approach.

and reduced effective star detection, remains a critical challenge for aerospace, ballistic, and marine applications. This review systematically analyzed the mechanisms of accuracy degradation and categorizes existing improvement methods into two paradigms: active and passive deblurring methods and motion blur compensation methods. Key conclusions and insights are summarized as follows:

##### A. Active and passive deblurring methods:

- **Passive deblurring methods:** such as optical system enhancements (e.g., EMCCD/ICCD sensors, large-aperture lenses) and multi-FOV designs (e.g., HYDRA), focus on reducing motion blur during imaging through hardware optimization. These approaches achieve detectable angular velocities up to  $4^\circ/\text{s}$  but face limitations in cost, size, and adaptability to complex motion.
- **Active deblurring methods:** including servo tracking platforms (e.g., NAS-27) and TDI/DBA techniques, actively counteract motion effects through mechanical isolation or charge synchronization. While effective for unidirectional motion, they struggle with multi-axis dynamics and add hardware complexity.

**Table 1**  
Methods to improve the attitude accuracy of the star sensor under dynamic conditions.

| Method category                  | Method description                                   | Advantages                                                                 | Disadvantages                                                                                | Examples and attitude accuracy under dynamic conditions                                                         |
|----------------------------------|------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Passive deblurring methods       | Improving the performance of optical imaging systems | Improved detection sensitivity, shortens star image exposure time          | Limited by material and manufacturing processes                                              | HAST, star recognizable <4°/s; Beihang University with its EMCCD and ICCD star sensor product                   |
|                                  | Multi-FOV star sensors                               | Increased FOV and number of efficient star images                          | Increases device volume, mass, and cost, special lens design required and calibration needed | HYDRA, star recognizable <10°/s; Tsinghua University                                                            |
| Active deblurring methods        | Servo tracking platform techniques                   | Reduced motion effect of multiple directions                               | Large volume, complex structure                                                              | Northrop Company                                                                                                |
|                                  | TDI and DBA                                          | Reduced motion effect of one (TDI, two for DBA) direction(s)               | Ineffective for complex motion                                                               | AST-301, 0.57 arcsecs (x, y), 19 arcsecs (z) @30°/s star recognizable <2.1°/s                                   |
| Motion blur compensation methods | Restoration of motion-blurred star image             | Eliminated complex motion blur                                             | Noise and error of motion parameters estimation                                              | Beihang University, Tsinghua University and etc., not available                                                 |
|                                  | Star centroiding with filtering method               | Improved centroiding accuracy using predicted centroids                    | Requires high-precision gyro measurement units                                               | star centroiding accuracy up to 1/1000 pixel in simulation reported by Boeing Company                           |
|                                  | ACF and ACFA                                         | Eliminated the impact of random noise, enhanced reliability and robustness | Requires high-precision gyro measurement units                                               | National University of Defense Technology, accuracy improvement proportional to the number of correlated frames |

### B. Motion blur compensation methods advancements:

- **Algorithmic innovation:** Motion-blurred star image restoration (e.g., RLA, Wiener filtering), centroid filtering (e.g., Kalman fusion), and the ACF/ACFA approaches demonstrate significant potential to enhance SNR and centroid accuracy. These software-driven solutions offer flexibility and cost efficiency compared to hardware modifications.
- **Multi-Frame correlation:** The ACF method, leveraging sequential frame correlation via strapdown gyroscopes, achieves virtual FOV expansion and noise suppression, improving attitude accuracy proportionally to  $1/\sqrt{N}$  (number of correlated frames). This approach mirrors the benefits of multi-FOV systems through time-division multiplexing, offering a scalable alternative to spatial hardware redundancy.

### C. Application-specific recommendations:

- **Spacecraft/satellites:** Multi-FOV star sensors (e.g., HYDRA) and ACF methods are ideal for high-reliability, long-duration missions.
- **Marine platforms:** Servo tracking systems and EMCCD-based sensors suit environments with moderate dynamics and relaxed size constraints.
- **High-dynamics scenarios:** ACF/ACFA methods paired with high-precision gyroscopes provide optimal accuracy under rapid maneuvers (e.g., missiles, agile satellites).

In summary, APDM methods address motion blur at the hardware level but face inherent trade-offs, while MBCM approaches provide flexible, cost-effective software solutions. The advantages and disadvantages of each method are described and summarized in Table 1. Mission-specific requirements — balancing precision, complexity, and operational constraints — should guide the selection of improvement strategies.

The CMOS APS is an ideal star sensor image sensor, and its detection sensitivity, stability, and other properties continue to improve. There is

a trend to replace CCD, the rolling shutter effect of CMOS star sensors under dynamic conditions are reported in [57–59], and CMOS APS star sensor dynamic performance enhancement technology is bound to be developed further. Star sensors are widely used inside the atmosphere, and the all-time performance requirement is a necessary consideration. All-time star sensors are a current and future research hot spot [60–63]. In addition, with the performance improvement of hardware of the star sensor and the extended application scenarios, the motion blur compensation methods would be complicated and varied [64,65].

### CRediT authorship contribution statement

**Liheng Ma:** Writing – original draft, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Dongkai Dai:** Writing – review & editing, Visualization, Validation, Software, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Yuanman Ni:** Writing – review & editing, Visualization, Validation, Software, Resources, Formal analysis.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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### Data availability

The data that support the findings of this study are available in the article.

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